

Microbial Control of the Emerald Ash Borer (Coleoptera: Buprestidae) Using *Beauveria bassiana* (Hypocreales: Cordycipitaceae) by the Means of an Autodissemination Device

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Abstract

The FraxiProtec, an autodissemination device loaded with the fungus *Beauveria bassiana* isolate CFL-A, was tested in the field to evaluate its potential to infect emerald ash borer adults and reduce their populations. During the 2-yr experimental period, the dispersion of *B. bassiana*-infected adults was also documented to assess the dissemination capacity of the biocontrol agent beyond the treated areas. The mean percentage of infected emerald ash borer in 2017 and 2018 in 15 treated sites was $43.3 \pm 2.9\%$ and $39.7 \pm 3.9\%$, respectively, and no significant variation was observed over the tested years. Furthermore, a 40% significant reduction of the mean emerald ash borer population growth per tree in treated sites was recorded when compared to the control sites. Emerald ash borer infected with *B. bassiana* isolate CFL-A were collected from baited sticky traps in the sentinel stations in the area surrounding the FraxiProtec-treated sites. Even at a distance of 125 m, an average of $9.4 \pm 2.6\%$ infected emerald ash borer were collected. Finally, exploratory analyzes were carried out on parameters such as the area to be treated, ash tree density, and FraxiProtec density to document potential relationships, which could be used in the determination of a prescription.

Key words: emerald ash borer, *Agrilus planipennis*, *Beauveria bassiana*, biological control, FraxiProtec

The emerald ash borer, *Agrilus planipennis* Fairmaire (Coleoptera: Buprestidae), is a phloem-feeding beetle native to Asia that attacks healthy ash trees (*Fraxinus* Linnaeus; Oleaceae) in North America. Ash species indigenous to eastern Asia share a coevolutionary history with emerald ash borer and present more resistance characteristics than evolutionarily naïve American hosts (Villari et al. 2016). In Asia, the emerald ash borer typically attack dying and stress host trees. Indeed, this pest caused little damage to three widely distributed ashes, *F. chinensis* Roxb, *F. rhynchophylla* L., and *F. mandshurica* Rupr. in China (Wei et al. 2007, Bauer et al. 2015), compared to the threatened ash species *F. pennsylvanica* Marsh. var. *lanceolata* (Borkh.) Sarg, which was introduced from North America in the early 1960s to eastern Asia (Wei et al. 2007).

In the absence of such coevolutionary history and resistant traits, the invasion of North America by emerald ash borer has caused the widespread mortality of ash trees, with devastating ecological and economic impacts (Herms and McCullough 2014, Villari et al. 2016). Since its accidental introduction in the early 1990s in southeast Michigan (Siegert et al. 2014), emerald ash borer populations are now established in 35 U.S. states and five Canadian provinces (McCullough 2019, CFIA 2020). Unfortunately, tactics to manage emerald ash borer are limited and difficult to evaluate, primarily because of the difficulty of detecting new infestations. However, several management strategies against emerald ash borer have been deployed in North America, which focus on classical biological

control and insecticide protection of high-value trees (Herms and McCullough 2014). Most of these control methods have targeted the immature stages of this phloem-feeding beetle. Ecological studies initiated in 2003 in northeast China revealed that natural enemies cause high mortality of emerald ash borer eggs and larvae, and play a critical role in suppressing emerald ash borer densities in forested areas of Asia (Duan et al. 2018). After the discovery of these Asiatic biocontrol agents, three hymenopteran parasitoids from China (*Tetrastichus planipennisi* Yang, *Spathius agrili* Ashmead, and *Oobius agrili* Zhang and Huang) were approved for introduction in the United States in 2007 following extensive host range testing and safety evaluation (Federal Register 2007, Bauer et al. 2015, Duan et al. 2015). In addition to classical biological control methods, there are insecticide products that are registered as options for protecting individual ash trees against emerald ash borer larval damage. Most insecticides used for emerald ash borer control act systemically and must, therefore, be translocated within the tree in order to provide protection of the plant's vascular system (Herms et al. 2014). Recent economic analyses have concluded that treating landscape ash trees with effective systemic insecticides is much less costly than removing trees (Herms et al. 2014).

Entomopathogens offer an effective avenue for the control of the emerald ash borer. Bauer et al. (2011) reported evaluating the potential of the microbial insecticide Bt SDS-502 (Asano et al. 2003) to suppress emerald ash borer adult populations by aerial application. Laboratory studies have shown toxicity of Bt SDS-502 to emerald ash borer adults and also confirmed that this Bt strain is compatible with exotic and native emerald ash borer parasitoids (Bauer et al. 2011). Entomopathogenic fungi are also particularly attractive candidates for use in biocontrol since fungal diseases may cause mortality in emerald ash borer populations (Macquarrie and Scharbach 2015, Duan et al. 2018) and because they can cause infection at all life stages (Hajek and St. Leger 1994). The use of entomopathogenic fungi also needs to be considered as a potential management strategy. More specifically, studies conducted in recent decades have demonstrated the entomopathogenic virulence of different isolates of *Beauveria bassiana* (Balsamo) Vuillemin, most of which were indigenous to infested emerald ash borer areas (Liu and Bauer 2006, 2008; Castrillo et al. 2010; Johny et al. 2012; Lyons et al. 2012). According to these observations, *B. bassiana* appears to be a promising biocontrol agent for regulating emerald ash borer populations.

Based on many years of laboratory isolates screening and following several field experiments, Lyons et al. (2012) have shown the potential of the entomopathogenic fungus *B. bassiana* as a microbial pesticide for the management of emerald ash borer populations (Lyons et al. 2012). The isolate CFL-A of *B. bassiana* (also known as INRS-CFL and INRS-242) was tested during these studies, and its effectiveness against emerald ash borer adults was demonstrated (Lyons et al. 2012). This isolate is an endogenous fungus recovered during field surveys from the pine shoot beetle, *Tomicus piniperda* L., near Cookshire, QC, Canada (Srei et al. 2020).

In this paper, we present the latest results of *B. bassiana* isolate CFL-A, which was tested in an autodissemination device (ADD) provided with a home-made contamination chamber. This ADD named FraxiProtec was developed to contaminate adults *A. planipennis* with *B. bassiana* isolate CFL-A conidia, and allow adults to disperse the fungus to their conspecifics during sexual feeding maturation and mating after emerald ash borer exited the trap. The horizontal contamination was demonstrated by Srei et al. (2020). This strategy, first introduced by Ignoffo (1978), is a known method for the control of pest insects. It has been tested on many coleopteran species like *Otiorhynchus sulcatus* Fab. (Pope et al. 2018), *Rhynchophorus*

ferrugineus Oliv. (Dembilio et al. 2018), *Dendroctonus simplex* LeConte (Srei 2017, Srei et al. 2017), *Hypothenemus hampei* Ferrari (Mota et al. 2017), and *Pachnoda interrupta* Oliv. (Habtegebreiel et al. 2016). However, in the FraxiProtec device, a fungal pouch with living hyphal with conidia is used rather than a load of dry conidia powder only (Lyons et al. 2012).

The hypothesis related to this project is that the use of FraxiProtec loaded with *B. bassiana* can significantly impact emerald ash borer adult mortality in treated sites, compared to untreated populations. Our objectives were 1) to document the average rate of emerald ash borer adult fungal infection in FraxiProtec-tested sites, 2) to evaluate the level of captures after a 2-yr experimental period in treated and untreated trees, and 3) to document the dispersion of *B. bassiana*-infected adults beyond the treated areas, to assess the dissemination capacity of the biocontrol agent.

Materials and Methods

FraxiProtec Specifications and Fungal Production

FraxiProtec pouches (14.5 × 11 cm; fiberglass mosquito net), as described by Srei (2017), are filled with sterile pearled barley in order to provide a growth medium to *B. bassiana* (Fig. 1). Before its use, the isolate CFL-A of *B. bassiana* was stored in 70% glycerol and maintained at -80°C in the Fungal bank of the INRS-Armand-Frappier Santé Biotechnologie, Laval, QC, Canada. The CFL-A isolate was first grown in Yeast Extract Peptone Dextrose broth (YPD, Alpha Bioscience Inc., Baltimore, MD), in a rotary shaker (150 rpm) for 4 d at 25°C. The sterile barley pouches were then inoculated with the fungal conidial suspension, before incubation for 10 d in the growth chamber under controlled conditions (25°C, 70% humidity, and in complete darkness). At the end of this process, the weight of the FraxiProtec pouches is 151.9 ± 0.8 g (mean ± SE) and the average spore count was evaluated at 1.93 × 10⁹ spores per g. All the *B. bassiana* pouches produced were stored at -20°C, and its conidial viability was evaluated before their use in the ADD.

Efficacy of the ADD

Before the deployment of the ADD, FraxiProtec was introduced in autocontamination chambers (Fig. 1) developed by Lavallée and Guertin, as described in Lyons et al. (2012), which was adapted at the base of a green Lindgren multifunnel trap (Synergy Semiochemicals Corp., Burnaby, British Columbia, Canada). The green Lindgren multifunnel traps were used for their efficacy to catch the emerald ash borer adults (Francese et al. 2011). The studies for assessing the efficacy of the ADD were carried out from 2017 to 2019 on mature ash trees in 11 emerald ash borer-infested municipalities in Quebec, Canada. Every ADD was paired with a green prism sticky trap (Distributions Solida Inc., St-Ferréol-les-Neiges, Québec, Canada), which was used to trap emerald ash borer in the surrounding area of the ADD. These green sticky traps were baited with a kairomone that mimics the host foliar volatile, (3Z)-hexenol, and a female synthetic pheromone, (3Z)-lactone (Chemtica Internacional, Santo Domingo, Costa Rica) to favor emerald ash borer trapping success. The ADD and the sticky trap were placed in the same tree at an average distance of 2.3 m from each other. Both traps were hung with ropes (Hughes et al. 2014) in the first third of the tree canopy, oriented toward the South to obtain as much possible sunlight (Herms et al. 2014, Poland et al. 2015). A minimum of 15 pairs of FraxiProtec and trapping devices were randomly distributed in experimental

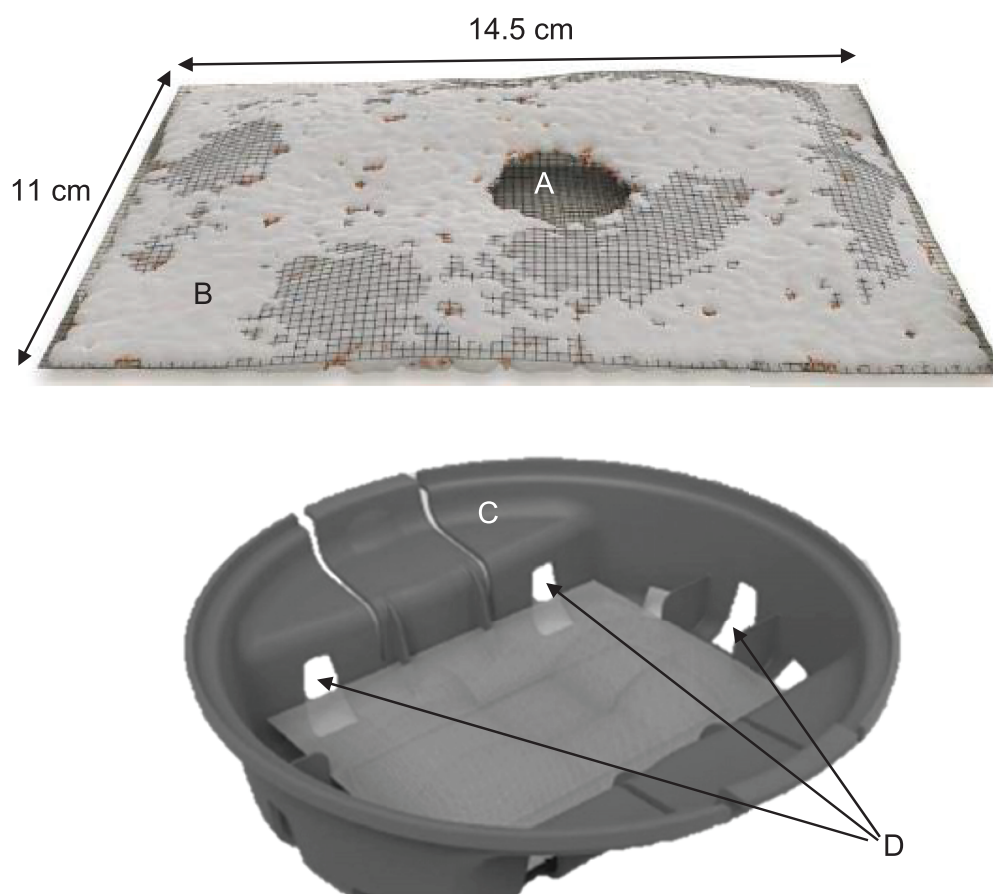


Fig. 1. Description of the FraxiProtec pouch and the contamination chamber. (A) FraxiProtec pouch. (B) Grown *B. bassiana* on the inoculated barley pouch. (C) Autocontamination chamber made to fit at the base of a Lindgren funnel trap. (D) Exit holes for contaminated emerald ash borer.

sites and installed in ash trees spaced by a minimal distance of 25 m from one another, to avoid interference of baited sticky traps between each other. The number of paired traps installed per treated site is presented in Table 1. The variation in the number of tested FraxiProtec per site was meant to provide information on the impact of FraxiProtec density per hectare on the fungal of emerald ash borer populations and also according to the ash tree density.

The FraxiProtec and green prism sticky traps were deployed at the beginning of the emerald ash borer flight period, corresponding to 250 degree-days (base 10°C), and for 4 wk. FraxiProtec pouches on which *B. bassiana* isolate CFL-A was grown were introduced in the contamination chambers on the ADD installation day and replaced after 2 wk to ensure an optimal conidia growth and viability. Weekly, insects trapped on green prism sticky traps were counted and individually collected in a 1.5-ml disposable clear centrifuge tubes (Fisher Scientific, Ottawa, Ontario, Canada) with sterilized tweezers. Collected emerald ash borer from each sampling period of each site were subsampled using the following Cochran's formula (Cochran 1977) to document the fungal infection rate:

$$n = (Z^2 \times p \times q \times N) / [(e^2(N - 1)) + (Z^2 \times p \times q)]$$

where n = estimated sample size; Z = level of error of 5% (set to 1.96); p = estimate of the true conversion rate in the population; $q = p - 1$; N = observed sample size; e = margin of error (confidence limit of 95%).

Data on insect population levels were collected in 2017 (baseline), 2018, and 2019, while data on fungal infection were recorded in 2017 and 2018.

Experimental Sites for Efficacy Tests in the Field

Experiments were conducted on mature ash trees randomly selected from street alignments, urban parks, and small woodlands in 11 emerald ash borer-infested municipalities in Quebec, Canada. Two control sites were paired with treated sites, from the same neighborhood, to document the impact of the treatment on the emerald ash borer population growth. Ash trees in control sites had to be at a minimum distance of 300 m from a site treated with FraxiProtec in order to limit bias from the migration of infected emerald ash borer. In these trees, green prism sticky traps were baited and installed, as previously mentioned. The detailed information on the treated and control sites and also according to presence of other control measures applied on these sites is presented in Tables 1 and 2. The size of ash trees was determined by measuring their diameter at breast height (DBH in cm) to validate the homogeneity of tree size among treatments.

Dispersion of *B. bassiana*-Infected Emerald Ash Borer Adults

The dispersion capacity of infected emerald ash borer adults was documented over the 2-yr experiment in the municipality of Contrecoeur, QC, Canada. Fifteen sentinel ash trees were selected at distances ranging from 25 to 125 m from the ADD experimental

Table 1. Localization and information on sites treated with *B. bassiana* isolate CFL-A

Municipality	Number of FraxiProtec	Site	TreeAzin treatment in site	Longitude	Latitude	Type of site	Ash species	Average ash DBH (cm)
Beaconsfield	15	Park Angell	Yes	-73.894130	45.428187	Urban woodland	White ash (80%)	30.5
							Red ash (20%)	27.2
Boucherville	15	Park Bois de Brouage and Broquerie	Yes	-73.448280	45.594579	Urban park	White ash (30%)	62.6
							Red ash (65%)	61.3
							Black ash (5%)	38.7
Boucherville	15	Fort St-Louis and Marie-Victorin Street	Yes	-73.456603	45.612039	Street alignment	Red ash (79%)	41.6
							Black ash (14%)	32.9
							White ash (7%)	61.7
Candiac	15	Park André-J.-Côté	Yes	-73.526776	45.399363	Urban park	White ash (13%)	37.6
							Red ash (87%)	39.7
Candiac	15	Park Fernand-Séguin	Yes	-73.509413	45.377027	Urban park	White ash (27%)	29.7
							Red ash (73%)	27.3
Contrecoeur	30	Park Cartier-Richard	No	-73.243004	45.855701	Urban park	White ash (28%)	53.5
							Red ash (72%)	44.7
Granby	15	Park Daniel-Johnson	Yes	-72.711178	45.400525	Urban park	Red ash (87%)	36.6
							Black ash (13%)	44.5
Laval	15	Centre de la Nature	Yes	-73.659811	45.605980	Urban park	White ash (73%)	26.3
							Red ash (27%)	28.8
Montréal South-West	15	Park Angrignon	Yes	-73.596599	45.445117	Urban park	White ash (7%)	48.9
							Red ash (86%)	50.8
							Black ash (7%)	29.9
Montréal	15	Park Mont-Royal_1	Yes	-73.597949	45.499905	Urban park	White ash (60%)	53.4
							Red ash (40%)	52.1
Montréal	15	Park Mont-Royal_2	Yes	-73.597000	45.499901	Urban park	White ash (38%)	51.2
							Red ash (62%)	41.9
Pointe-Claire	15	Park Hermitage	Yes	-73.838991	45.464975	Urban park	White ash (73%)	39.1
							Red ash (20%)	34.4
							Black ash (7%)	38.5
Repentigny	30	Park Premières-Nations	Yes	-73.430715	45.784921	Urban woodland	White ash (50%)	38.5
							Red ash (50%)	36.1
Rosemère	15	Park Bourbonnière	Yes	-73.806881	45.635959	Urban park	White ash (60%)	37.7
							Red ash (40%)	46.3
Rosemère	30	Park Rosenbloom	Yes	-73.795425	45.646500	Urban park	Red ash (100%)	31.7
							Global average	
							(±SE): 41.1 ± 0.9	

design located in the Cartier-Richard Park. In these ash trees, green prism sticky traps were baited and installed as previously mentioned to evaluate the capacity of emerald ash borer adults to disperse beyond the treated area once infected with the microbial agent. Emerald ash borer adults trapped on the sticky traps were collected over the same period of 4 wk, according to the methodology previously mentioned, and then submitted to a fungal infection investigation.

Validation of Fungal Infection

In the laboratory, a morphological observation and a molecular diagnostic were performed in order to validate the presence of the fungus *B. bassiana* CFL-A on emerald ash borer adults collected from each site. Insects were firstly placed in Petri dishes (245 × 245 × 25 mm, Thermo Scientific, Ottawa, Ontario, Canada) containing a 0.22-μm filter paper (Whatman International Ltd., Maidstone, United Kingdom) moistened with 10 ml of sterile osmosis water. The Petri dishes were then incubated for 14 d at 25°C, 60% humidity, and in complete darkness. The presence of a cottony white structure on the insect, the muscardine, was used as an indication of fungal infection by *B. bassiana* isolate CFL-A.

After incubation, emerald ash borer adults with muscardine were subsampled and molecular biological tools were applied as described by Srei (2017). Briefly, insects were individually crushed in 100 μl of phosphate-buffered saline (PBS: 137 mM NaCl, 2.7 mM KCl, 10 mM Na₂HPO₄, pH 7.4) containing 0.1% Triton X-100. Each sample was transferred into a 2-ml screw cap tube containing 200 mg of 0.1-mm glass beads (BioSpecs, Bartlesville, OK) for DNA extraction.

Total DNA was extracted using mechanical lysis method. For this, 1 ml of extraction buffer (50 mM Tris-HCl, 5 mM EDTA-2Na, 3% SDS, pH 8.0) containing 20 mg/μl of RNase A was added to each crushed sample. Cell lysis was performed in two cycles of 6 m/s for 60 s using the FastPrep-24 Instrument (MP Biomedicals, Solon, OH). The tubes were then centrifuged at 13,000 rpm for 10 min, and the supernatant was recovered before the second lysis cycle. Ammonium acetate was added to each supernatant at a final concentration of 2 M. Tubes were briefly agitated by inversion and left on ice for 5 min before centrifugation at 14,000 rpm for 15 min at 4°C. The supernatant was then recovered and placed on ice for 5 min before a second centrifugation under the same conditions. After adding an equal volume of isopropanol to the recovered supernatant,

Table 2. Localization and information on untreated sites and dispersion study site (*)

Municipality	Number of ash trees	Site	TreeAzin treatment in site	Longitude	Latitude	Type of site	Ash species	Average ash DBH (cm)
Candiac	15	Park Fernand-Séguin	Yes	-73.244029	45.853851	Urban park	Red ash (100%)	25.8
Contrecoeur*	15	In the streets around Park Cartier-Richard	No	-73.243004	45.855701	Street alignment	Red ash (100%)	65.3
Rosemère	15	Hardy Street	Yes	-73.782872	45.648525	Street alignment	Red ash (100%)	33.4
Global average (±SE): 41.5 ± 2.7								

tubes were placed overnight at 4°C to allow the precipitation of DNA. After centrifugation at 14,000 rpm for 30 min at 4°C, the supernatant was removed. The final DNA pellet was obtained by rinsing twice with cold ethanol 70%. Lastly, DNA pellet were air-dried and resuspended in 50 µl of sterile distilled water. DNA concentration was quantified using Nanodrop (ND-1000 Spectrophotometer, Thermo Fisher Scientific Inc., Wilmington, DE).

The B locus intergenic region was amplified using specific primers CF1B1F2 (5'-AAAAGGGACAATGGCTTGTA-3') and INRSRO (5'-TTGAGTTGGGCGAACTGATGCCCTCTT-3') to detect the presence of the fungus *B. bassiana* strain CFL-A. These primers were designed and graciously provided by Dr. Kyei-Poku's laboratory. All PCR reactions were carried out in 50 µl reaction volumes using 50 mM MgCl₂, 20 µg of N + BSA, 10 mM dNTPs, 10 mM of each primer, 5 U Taq DNA polymerase, 10× ThermiPol buffer (New England Biolabs, Whitby, ON, Canada) on a thermal cycler equipment. An initial denaturation of 3 min at 94°C was followed by amplification for 40 cycles at the following conditions: 94°C for 30 s, 62°C for 30 s, and 72°C for 1 min. The protocol was completed with a final 10-min extension at 72°C. The PCR products were confirmed using electrophoresis with 2% agarose gels stained with Gel Stain and visualized under UV light.

Statistical Analysis

All statistical analyses were performed using the statistical software package R, version 3.4.4 (R Core Team 2018). Statistical analysis was performed on tree size among the treated and control sites since it could have induced bias in this study. For this purpose, the tree DBH was subjected to an analysis of variance (ANOVA) in order to validate the tree size homogeneity among treatments. DBH data were submitted to a LOG transformation to respect the postulates of normality and homogeneity of variance. ANOVA was performed to determine the consistency of fungal contamination of the device in 2017 and 2018. Analyses on the percentage of fungal infection data were performed using a Generalized Linear Mixed Model (GLMM) with binomial distribution. Since the emerald ash borer population growth results (trapping data) did not conform to the distribution or homogeneity of variance, even with transformation, these data were subjected to the nonparametric Wilcoxon rank-sum test. A linear regression analysis was performed on the percentage of emerald ash borer infected according to the distance of the sentinel station where they were collected, measured from the treated site. Finally, regression analyses were performed on variables in study sites, such as FraxiProtec density per hectare, and ash tree density per hectare, in order to identify the most significant model that would fit with our fungal infection results. For all statistical tests, the level of rejection was set at $\alpha = 0.05$.

Results

The first analysis performed to document the site's homogeneity revealed that there was no significant difference in tree size, as measured by DBH, between treated and control sites ($F = 1.569$; $df = 237$; $P = 0.211$). This analysis confirms that the randomly selected trees of both treated and control sites are equivalent in size and should not cause any bias since it is known that trees with smaller diameter produced significantly fewer emerald ash borer than larger trees (McCullough and Siegert 2007).

For all the results presented in the following sections, the first-week data set of collected emerald ash borer on prism sticky traps was the only one used in the analysis and data interpretation. For both years of this experimental field study, failure was noticed in the capacity of the sticky traps to maintain its efficacy in trapping emerald ash borer over the 4-wk period, what greatly impacted the level of emerald ash borer catches and emerald ash borer infection level. For this reason, the data used were obtained while the sticky traps were fully operational, to provide the most appropriate interpretation of these experiments.

Fungal Infection of Emerald Ash Borer Adults

Emerald ash borer collected from sticky prism traps allowed us to quantify the percentage of infected adults that were exposed to the FraxiProtec. The mean percentage of infected emerald ash borer in 2017 and 2018 in treated sites was $43.3 \pm 2.9\%$ and $39.7 \pm 3.9\%$, respectively, with ranges of 23.3–77.6%, over the 2-yr study. No significant variation was observed in the percentage of infected emerald ash borer adults over the tested years ($F = 0.443$; $df = 20$; $P = 0.513$). In addition, since ash tree species were represented at different ratios in the treated sites, the statistical analysis demonstrated that ash species did not significantly impact on fungal infection results ($F = 0.118$; $df = 226$; $P = 0.732$).

Emerald Ash Borer Population Growth in Treated and Control Sites

The measures on emerald ash borer population levels were taken at the tree level at the beginning of the experimental period in 2017 (baseline) and in 2019, after 2 yr of treatment with FraxiProtec. Only the experimental sites in Candiac and Rosemère were subjected to this analysis since they included treated and control sites, in which ash trees were under TreeAzin treatments. The evolution of captures over the years, according to treatments, is presented in Fig. 2. It is possible to visualize the significant impact of the FraxiProtec treatment on emerald ash borer population after a 2-yr period, while the average initial population per tree, as documented by trapping results, was similar for both treatments (Fig. 2). Additional analyses were performed on the absolute

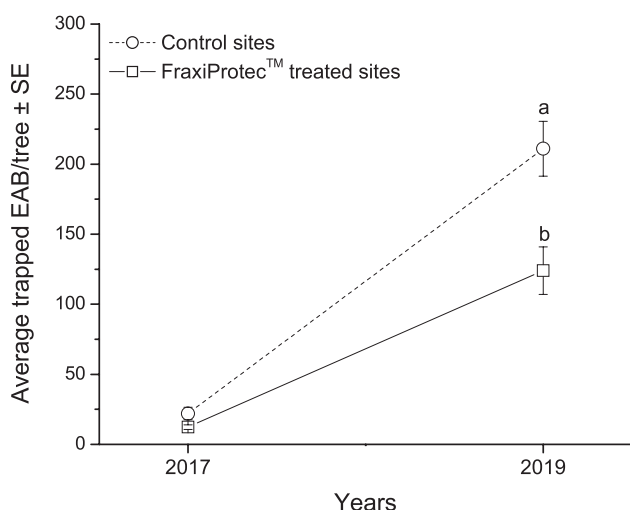


Fig. 2. Emerald ash borer trapping results per tree in 2017 and 2019 in control and FraxiProtec-treated sites. Curves with different letters are significantly different ($P < 0.05$).

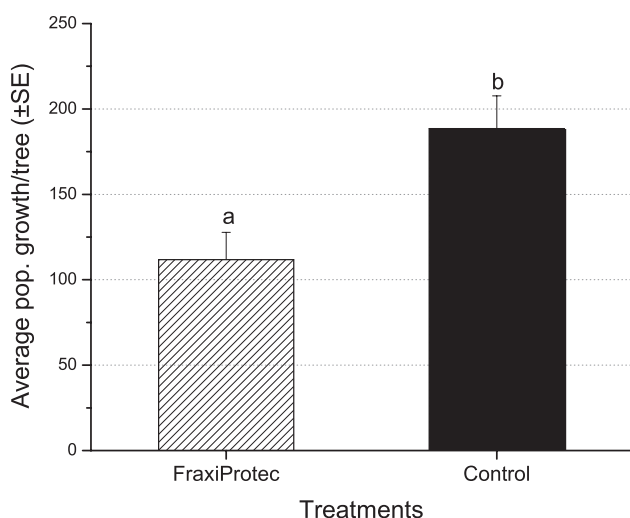


Fig. 3. Average population growth per tree in control and FraxiProtec-treated sites. Columns with different letters are significantly different ($P < 0.05$).

emerald ash borer population growth per tree, which means the number of emerald ash borer trapped on trees in 2019, minus the number of emerald ash borer trapped at the baseline in 2017, over a 1-wk period. The mean emerald ash borer population growth per tree in treated and control sites after a 2-yr period was $111.8 \pm 15.9\%$ and $188.6 \pm 19.0\%$, respectively (Fig. 3). Significantly lower population growth was observed in FraxiProtec-treated sites compared to the control sites ($W = 318.5$; $P < 0.001$). Indeed, the emerald ash borer population increase in the control sites was 40% more important than what was observed in the treated sites.

Dispersion of *B. bassiana*-Infected Emerald Ash Borer Adults

Infected emerald ash borer with *B. bassiana* isolate CFL-A were collected from baited sticky traps in the sentinel stations in the area surrounding the FraxiProtec-treated site, in Contrecoeur. A linear regression performed on the mean percentage of infected emerald ash borer adults demonstrated a significant negative relationship

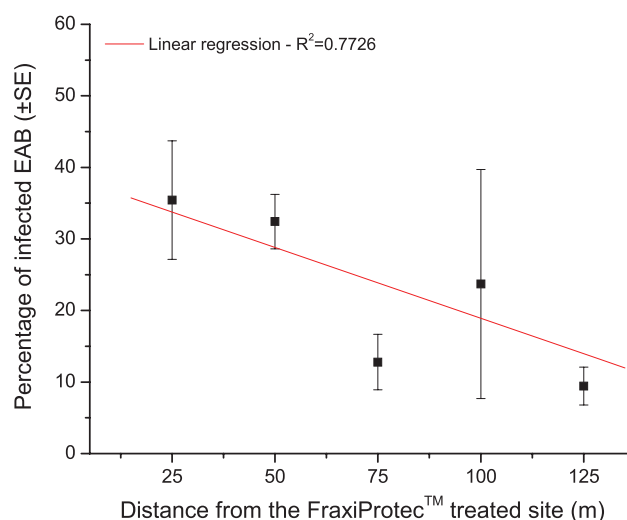


Fig. 4. Representation of the percentage of infected emerald ash borer results obtained in the dispersion study as affected by the distance from a FraxiProtec-treated site.

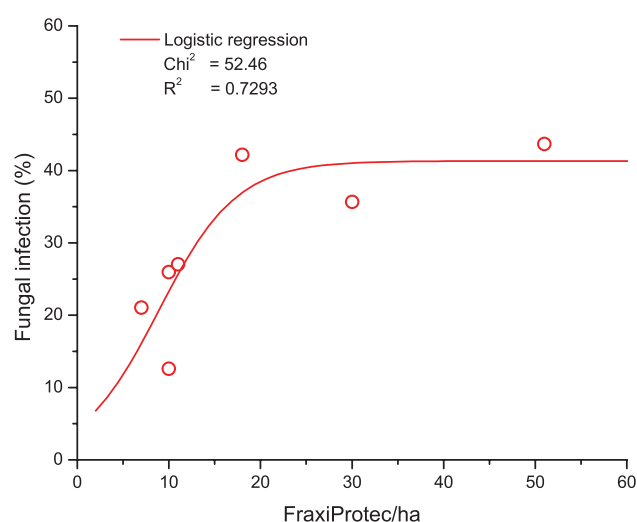


Fig. 5. Impact of the FraxiProtec density per hectare on the percentage of fungal infection fitted with a logistic regression model.

between the infection rate and the distance from the treated site ($R^2 = 0.7726$; $P = 0.0470$) (Fig. 4). Even at a distance of 125 m, an average of $9.4 \pm 2.6\%$ infected emerald ash borer were collected, which demonstrates that emerald ash borer can disperse on interesting distances following their contamination in the ADD.

Impact of Study Sites on FraxiProtec Efficacy

The heterogeneity of the study sites in the different municipalities regarding the area of parks and ash tree density allowed complementary analysis to document the potential of FraxiProtec to infect emerald ash borer adults in variable environments. An analysis performed between the percentage of infected emerald ash borer adults and the density of FraxiProtec per hectare shown a logistical regression relationship ($R^2 = 0.7293$) (Fig. 5). Furthermore, when the percentage of infected emerald ash borer was tested with the density of ash trees, another significant logistic fitted model was observed ($R^2 = 0.9268$) (Fig. 6). Finally, an exponential regression model was

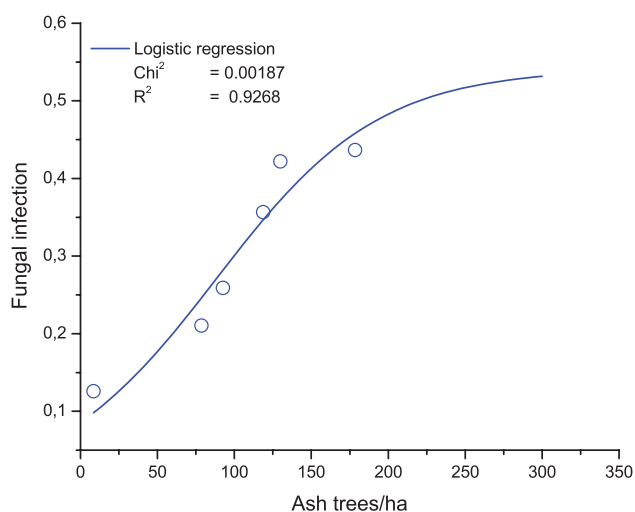


Fig. 6. Impact of the ash tree density per hectare on the fungal infection fitted with a logistic regression model.

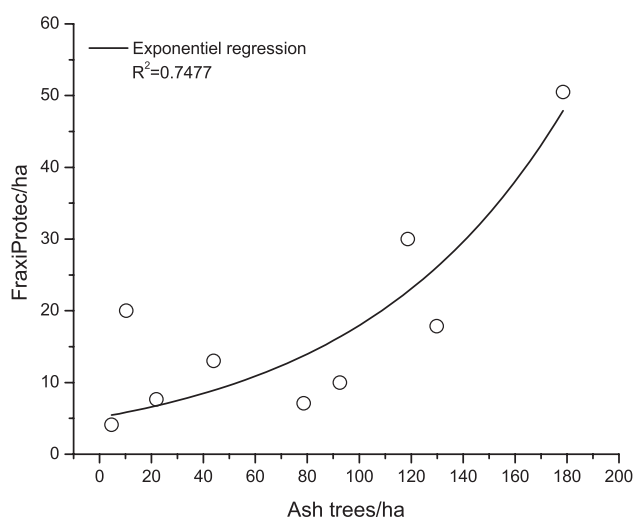


Fig. 7. Impact of the ash tree density on the needed FraxiProtec per hectare fitted with an exponential regression model.

obtained when the density of FraxiProtec was tested against the density of ash trees ($R^2 = 0.7477$) (Fig. 7).

Discussion

For the first time, an entomopathogenic fungus isolate was used under outdoor conditions in Canada against an invasive insect. The utilization of FraxiProtec in ash trees of parks or street alignments revealed new information on the impact of this biological control method on the emerald ash borer population management. The fungus *B. bassiana* isolate CFL-A demonstrated to be an efficient entomopathogen that can infect and kill emerald ash borer adults (Lyons et al. 2012, Srei et al. 2020) and also disseminate among the insect pest population. According to Lyons et al. (2012), the average conidia load on emerald ash borer after an exposure to FraxiProtec through the ADD was $579,200 \pm 86,181$ conidia per beetle. During their experiment, the median survival times for CFL-A (named INRS-CFL in Lyons' paper)-treated emerald ash borer ranged from

6 to 8 d and mortality of the exposed group to this isolate reached 100% within the duration of the experiment (Lyons et al. 2012).

This 2-yr period field experiment allowed us to validate its potential to infect adults and also its constancy over the years. Under experimental field conditions, an average of 41.5% of emerald ash borer infection was reached for both years, even if optimal parameters on using FraxiProtec were not still identified. This is much higher than the 2% reported by Bauer et al. (2004), who observed five species of fungus in Michigan but lower than the 58–83% reported by Liu and Bauer (2008) following field spray with *Beauveria* strain GHA. Exploratory studies on this autodissemination strategy using FraxiProtec have been carried out by Lyons et al. (2016), where they used fluorescent powders for tracking autocontamination of adult emerald ash borer with the entomopathogenic fungus, *B. bassiana*. An average of 8% of emerald ash borer trapped were positive for fluorescent powder and were found to be carrying viable conidia of *B. bassiana* isolate CFL-A using unbaited prism sticky traps.

The autocontamination strategy was also found to be an interesting avenue against the coffee berry borer, *H. hampei* (Ferrari). A trap was developed by Mota et al. (2017) to attract colonizing *H. hampei* females, contaminate them with *B. bassiana* conidia, and disperse the fungus to the environment after the beetles exited the trap. During their experiments, the total mortality of insects in autoinoculation traps with the fungus ranged from 66.3 to 100%, which was significantly greater than that from traps without the fungus (Mota et al. 2017).

The impact of this deadly infection caused by *B. bassiana* isolate CFL-A has also been noticed on the level of trapping results in treated and control sites. Indeed, the higher adult mortality in treated sites affected emerald ash borer population growth, what was reflected in the field by a reduction of the number of adults captured per tree. Even if ash trees in control sites were treated with TreeAzin, the impact of this treatment did not allow as much population reduction than the combined effect of FraxiProtec and TreeAzin. According to Jennings et al. (2015), there is approximately 5% of adult mortality caused from disease and predation, which is really low compared to immature stages mortality, which corresponds to more than 90%. However, the deployment of FraxiProtec in different ash tree settings favors greater adult mortality, which indirectly impacted on reproduction and oviposition.

The detection of emerald ash borer-infected adults trapped in the sentinel trees, during the dispersion experiment, demonstrated that adult flight and dispersal are possible following adult contamination even with a heavy conidia load in the ADD (Lyons et al. 2012). This behavior is interesting since it means that these individuals could be vectors of the pathogen in the surrounding area from where the FraxiProtec is installed. This phenomenon is well-documented since a wide variety of fungal species stand alone in the extent and complexity of their manipulation of arthropod behaviors (Lovett and St. Leger 2017). Horizontal transmission of *B. bassiana* isolate CFL-A was investigated using emerald ash borer as a vector during a mating laboratory experiment (Srei et al. 2020). The results showed that male-to-female transmission of the fungal isolate CFL-A (identified as INRS-242 in Srei et al. paper) within emerald ash borer adults was higher than from female-to-male, what might be explained by the presence of pubescence of emerald ash borer males. It was also noticed in a preliminary experiment that some emerald ash borer males can mate with at least two different females (Srei et al. 2020). Dispersion of infected individuals, horizontal transmission, and multiple mating behavior in emerald ash borer males are key elements for optimizing the dissemination of *B. bassiana* and favor the control of this invasive pest population.

Considering the unknown dispersal and density of emerald ash borer population in a specific area, the identification of the appropriate prescription for the FraxiProtec is not an easy task. However, it is a prerequisite in order to reach the optimal infection level of emerald ash borer population management success. The exploratory analyzes conducted on key parameters, such as area to be treated, ash tree density, and FraxiProtec density, revealed interesting relationships. For instance, the analysis of the success of the ADD to contaminate emerald ash borer adults using a logistic regression model predicted the level of fungal infection on the basis of FraxiProtec density per hectare. The density of ash trees also appears to be an expected characteristic to consider since it was correlated to FraxiProtec density and the level of fungal infection observed. The interpretation of these models first reveals that the more trees there are in a given area, the more emerald ash borer adults might be present for that surface area. If there are several attacked trees within the area where FraxiProtec and the prism sticky traps are hanged, it would be probably normal to attract and contaminate more insects, because its radius of action might not be limited to the tree where it has been hung. Finally, using the exponential regression model allowed to generate an estimate of FraxiProtec required according to a given ash tree density per hectare, under the currently conducted field experiments.

Another important aspect that would be interesting to consider is the impact of this biological control strategy on ash tree capacity to survive, according to the intensity of treatment. The use of FraxiProtec impacts on the control of emerald ash borer adult populations and its success is modulated by previously identified parameters. However, the damage done to ash trees is caused by emerald ash borer larvae and this aspect would need to be documented according to treatment. Indeed, emerald ash borer larvae consume phloem tissue beneath the bark, creating feeding galleries that eventually cut off the transport of carbohydrates in the phloem tissue (Wang et al. 2010). Data collected by McCullough and Siegert (2007) showed that an average emerald ash borer density of 88.9 ± 4.6 beetles per m² of surface area was found to be sufficient to kill ash trees. Understanding the relationships among tree characteristics, site attributes, and tree survival would greatly facilitate the development of management strategies to protect ash stand health (Knight et al. 2013).

In conclusion, the deployment of FraxiProtec showed to have a significant impact on emerald ash borer population reduction, compared to the control sites. Relevant information obtained during these experiments on the potential and the extent of this control method will provide us with data for prescription determination and FraxiProtec implementation. In addition, further study sites were added in Illinois, Pennsylvania, and Minnesota over the last years to favor data collection on the potential of this ADD. Finally, additional studies will be required in order to enhance the efficacy of the FraxiProtec to further reduce emerald ash borer population growth and eventually limit emerald ash borer larval density at a threshold under which ash trees will be able to successfully establish their defense mechanisms.

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